

A Low-Cost Urban Search and Rescue Robot for Developing Countries

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Abstract—Since urban search and rescue is a difficult domain for autonomous mobile robots to operate in, the environment can be expected to be highly unstructured, with many obstacles and hazards for a robot to deal with. Besides, if human rescue teams are going to accept robotic assistance, they need to be assured that the robots are going to be helpful, not a hindrance. With these factors in mind, we work toward the development of a swarm of small, inexpensive search and rescue robots. The platform has been developed to be low in cost and easy to construct from commonly available parts so that they are suitable for developing countries like Bangladesh. The robot can provide useful information to human rescuers without requiring specialized knowledge in operating robots. It can be controlled remotely from a station. It sends continuous thermal and video feed, environmental sensory data of the disaster site to the associated web application through the control station. It can run through rough terrain in a completely dark environment with the help of a powerful remote-controlled flashlight. It can detect living humans inside the rubble and can establish communication between them and the rescuers.

Index Terms—Terrain-ability, Rescue Robot, Sensor package, thermal camera, Urban search

I. INTRODUCTION

When a disaster strikes, such as an earthquake, hurricane, typhoon, flood, dam failure, technological accident, fire, collapsed building due to natural or man-made events like - terrorist activities or the release of hazardous materials, rescue workers have to step in, in order to prevent, or at least minimize, the loss of human lives. While the situation can vary a lot, the main objective of the rescue workers is to find humans in need of assistance (Search) and to move them out of harm's way (Rescue).

Though humans can handle any assignment and a huge variety of tasks after a short training period, there are several risks and disadvantages involved when working in a search and rescue task force :

- Risk of physical injury, such as cuts, scrapes, burns, and broken bones.
- Risk of respiratory injuries due to hazardous materials, fumes, dust, and carbon monoxide.
- Risk of diseases such as diphtheria, tetanus, and pneumonia.
- Risk of psychological and emotional trauma caused by gruesome scenes.
- Humans are not expendable.
- Humans are prone to make mistakes.

As we want to provide primary life support of victims in any disaster and have to keep the rescue team safe, the best solution at hand is a compromise where humans and robots team-up. Where the robot does repetitive, dull, hazardous and dangerous parts of the work, while the human concentrates on finding patterns, making conclusions and helping the robot to understand what part of a task to concentrate on and in what order to execute its sub-goals. The general idea is to let the robot do things it does better than the human, and vice versa. Rescuing survivors from a collapsed building has proven to be one of the toughest challenges in today's world. But in the context of a rescue team, it is even more challenging to detect the specific regions of a large building and identify victims. Moreover, the overall environment of Urban Search and Rescue (USAR) does not give itself a speedy survey and retrieval process. The terrain is extremely flustered and the regions for exploration are often irregular and very constrained. And these problems always make human rescue teams difficult to accomplish their tasks. If they have prior knowledge about the terrain's internal structure and environment, they can effectively deploy rescue operations in specific regions of the whole building. In this work, we designed a robot that can be controlled from a control station remotely. The task of the robot is to send continuous video feed, environmental sensory data of the building to the control station. These fetched data will help the human rescue team to plan their operation adequately. And above all the robot will be able to detect living humans who are obstructed inside the building and their positions which will increase the retrieval performance of the rescue team.

The robot internal architecture of the robot consists of a control unit, a mechanical unit, and an application unit. The control unit stores programs that process the commands from the control station and thus handles the communication of the robot. The control unit is also responsible for fetching environmental parameters. The mechanical unit consists of motors for the movement of the robot and a very capable sensor package which includes cameras, a gas sensor, an altitude sensor, a thermal sensor, etc.

A. Usefulness in the Context of Bangladesh

Since Bangladesh is a very dense country and a country of natural hazards, we should be proactive about any upcoming disaster. If we have some rescue robots, then we can search and

rescue human at any hazardous site faster and more efficiently and can save more life. Few recent incidents in this sub-continent where this kind of rescue robots can come to play are listed below:

- **Earthquake Risk in Bangladesh** - Recently scientists have come to recognize that Bangladesh is positioned at the juncture of several active tectonic plate boundaries [1], [2]. Moreover, it sits atop the world's largest river delta at close to sea level, facing both the risk posed by a quake and secondary risks of tsunamis and flooding in the quakes aftermath. Rescuing survivors from a building collapsed due to earthquakes has proven to be one of the toughest challenges in today's world. But in the context of a rescue team, it is even more challenging to detect the specific regions of a large building and identify victims. The terrain is extremely cluttered and the regions for exploration are often irregular in nature and very constrained. And these problems always make human rescue teams difficult to accomplish their tasks. A low-cost search and rescue robot will help the human rescue team to plan their operation adequately.
- **Rana Plaza Incident:** Rana Plaza was a nine-story commercial building at Savar, Dhaka collapsed on 24th April 2013 [3]. More than 1134 garment workers died and several hundreds were missing in the collapse. It took 20 days to conclude the rescue efforts of this nine-story building whereas rescue efforts in The World Trade Centre took only 10 days to complete as they had a specialized team with robots for help and acceleration of the search task. If during the Rana Plaza incident, we have some rescue robots, we could have completed the search task with less time & thus can save more people.
- **Fire Hazard:** In March 2019, nine devastating hazards took place across Bangladesh including fire, lightning, hailstorm, storm, nor wester, building collapse, bridge collapse, boat capsize and wild animal attack. The most occurred hazard of this month was fire with 24 incidents in 11 districts correspondingly Dhaka, Chattogram, Madaripur, Chandpur, Rangpur, Lakshmipur, Cumilla, Jhalokathi, Gopalganj, Gazipur and Noakhali.
- **Toxic Gas Release:** Toxic gases (or noxious gases) are gases that are harmful to living things. They can easily build up in confined working spaces when the production process uses noxious gases. It may also result in the biological chemical breakdown of a substance that is being stored in a tank. Certain factory-related activities, such as welding, can also result in the build-up of toxic gases in a confined space. Our design includes gas and thermal sensors to effectively identify hazardous gases that are harmful to living things and notifies the operators immediately.
- **Children Stuck in Deep-hole:** A two-year-old Indian boy stuck in a narrow well for more than four days was pulled out dead, triggering protests over delays in reaching the toddler. The case captured national attention after he fell

into the 33-meter (110 feet) deep well in the Punjab state district of Sangrur while playing. His family was not sure whether he fell into the well and concluded that the rescue operation was "delayed due to lack of required technical assistance". Search and rescue robots of smaller size and greater range can work as a communication aid between a trapped victim and the rescue team.

An urban search and rescue robot could accelerate the rescue operations of the above-mentioned situations. In developing such an urban search and rescue robot, our main contributions are as follows:

- The USAR can be sent inside a collapsed building to find humans trapped inside using thermal imaging
- A web interface has been developed so that anyone from anywhere in the world can get a live update of the scenario
- The rescuers can get an overall idea of the internal environment and take necessary measures before going in
- The USAR is equipped with a powerful flashlight, thus it can send clear videos even in complete darkness
- The USAR can traverse through the debris with ease, even 60-degree inclination
- Once the bot reaches to the imperiled, the rescuers can converse with them via the robot

II. RELATED WORKS

The very first considerations before starting to build a search and rescue bot are to ensure the key features like mobility, agility and high accuracy based map generation capability of the robot. Sheh et al. [4], for example, was highly concerned about the escalation of the mobility of a robot. A toy "Tarantula" was customized and reshaped to be a rescue robot called "The Redback." The robot had enhanced and upgraded mobility features because of its small size and lightweight. Thrun and his teammates [5] introduced the Fast-SLAM algorithm (A Factored Solution to the Simultaneous Localization and Mapping Problem) to support increasing its accuracy in map generation. Rescue robots sometimes work as a group. They have the power to communicate with each other and make certain decisions so effectively that as a whole help the rescue team to strengthen their operation. Vargas and his co-workers [6] constructed three interdependent mobile robots that were capable of detecting adversities in disasters. In another point of view, Birk and his colleagues [7] built a rescue robot to survey a collapsing building in urban localities and find victims. Whether Vincent and Trentini [8] introduced a robot that was capable of interpreting obstacle shapes, sizes and was able to climb over them rather than avoid them by implementing image segmentation and various shape detection algorithms. In contrast with manually controlled robots, autonomous rescue robots must have the ability to decide their travel paths in many circumstances which direct them to traverse the collapsed area comfortably and gain the information about certain environmental parameters and victims. This requirement guides most of the researchers to

implement advanced technologies for getting intelligent terrain robots. Birk and Kenn [9], for instance, have developed a semi-autonomous rescue robot to run-over the drawbacks of wireless communication that blocks constant control over the robot. Besides, researchers have developed a duel mode rescue robot to perform in either fully autonomous or teleoperation mode [10], [11]. [12], [13] and path planning [14], [15]

In Bangladesh, the attempt to build USAR is not completely novel. In 2013, GK Dey and his teammates built a mobile robot equipped with a metal detector, gas and smoke sensor, ultrasonic sensor, video camera, and a robotic gripper [16]. But it didn't have any system for human detection. In 2015, Chowdhury et al., developed a USAR focusing on strengthening the signal for uninterrupted communication using wifi beacons [17]. In the same year, HU Zaman and his team developed a USAR equipped with temperature, humidity, and gas sensor using the Bluetooth module [18]. In 2016, an amphibious smart USAR was introduced by MN Jiko and his team which could climb stairs and also floats on water [19]. The robot was equipped with several environment sensors and could be controlled from anywhere via a web server. This robot could be very efficient for surveillance and reconnaissance. But as it couldn't detect living beings, it was not suitable for rescue missions. In the same year, a semi-autonomous rescue robot was developed by Uddin et al., which used a PIR motion sensor for life detection [20]. But the major flaw of this method was, the human trapped in a disaster site could be unconscious or unable to move. In that case, imperiled would stay undetected by the robot. Our USAR aims to solve this problem using a low cost 8x8 GRIDEYE thermal camera and also send several environmental sensory data through wireless communication.

III. OUR APPROACH

Mobile robots designed for search and rescue operations are rugged in design and offer many features to address the above constraints. Features considering the robot are listed below:

- Tele-operatable Robot - The robot could be deployed at any point in the site while the operator stays safely outside the perimeter of the ruin.
- Audio - It can transmit an audio signal from the site from which also helps the operator to know if there are any people alive and make conversations with them.
- Visual - There are several types of cameras such as - wide-angle camera (FPV camera), a thermal camera located at different positions of the robot to work in different environments. From the video feeds operator can easily find humans by precisely. After finding human, it will send his/her altitude to the rescue team.
- Altitude - It gives the altitude of any human found inside the debris. This will be very helpful for the rescue team to find out the floor if the disastrous building is some multi-storied large building.
- Website - There is a website that shows the two video feeds (thermal camera and FPV camera feed) and the environmental parameters (altitude, temperature, amount

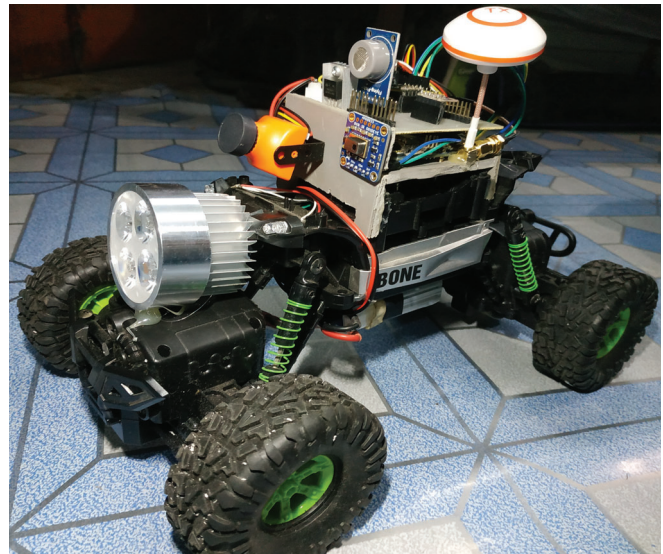


Figure 1: Physical structure of the bot

of CO gas around the robot). Anyone from any kind of device with internet connection can easily watch online what is happening around the robot.

- Power Efficiency - Batteries on the robot can last for at least a 30 minute run time.

IV. DESIGN PRINCIPLE

In real life operations of such search and rescue robots, due to a highly unstructured environment, it is very possible for the robots to fall into pits or holes, get stuck in cramped spaces or lose communication. However, situations like these are likely to happen when a disaster site is searched using robots and these losses are part of the operational cost. Though large sophisticated units would indeed deliver better results, losing one unit during operation would cost a fortune. Therefore the cost of manufacturing of the robots should be kept as low as possible to receive the highest benefit. That is why our robot is carefully designed with sensors that will yield the best result while keeping the cost to a minimum.

The design of the robot can be divided into three sections:

A. Mechanical Components

The main platform of the bot containing all the electrical components is mounted on a body with 4-wheel drive and 4 independent suspension springs which is driven by radio control. It can climb slopes up to 60° and move easily on different terrain conditions. The main platform is made from a PVC board and mounted on the body with screws.

B. Electrical Components

The bot has multiple sensors onboard. The main microprocessor is an Arduino UNO.

- An Arduino shield has been designed for accommodating the sensors and modules.
- There is an MQ-7 Gas Sensor that gives different analog voltages in different amounts of Carbon Monoxide

present in the air. It will alert the rescuers about the places where the gas mask is needed.

- A BMP-280 Barometric sensor that communicates via I²C is used to determine the altitude of the bot, therefore, informing which floor the bot is on. It also gives the ambient temperature reading.
- An 8x8 Array Grid-EYE Thermal Camera Sensor is used that can detect humans within 2m range which also uses I²C communication.
- There's also a 4W LED flashlight that can be turned on/off remotely from the station.
- There is an FPV Wide-angle Camera and a microphone on-board.
- There is a small walkie talkie on-board which can communicate within 1 km with the other one at the station. Rescuers can communicate with the people in distress with the help of this.
- The wheels are driven by 4 high torque motors.
- There are two sources of power. One is an 11.1V 3 cell 1000 mAh Lithium Polymer rechargeable battery which is mounted underneath the bot. The other is a 6V 600 mAh rechargeable battery that is placed within the body. The power from the first one is split into two channels. One is regulated to 5V which powers up the Arduino and supplies power to the on-board sensors and camera. The other is the original 11.1V which powers up the LED flashlight and the Audio Video(AV) transmitter. The 600 mAh battery powers up the motors and the receiver for remote control operation. It lasts 20 minutes on heavy use.

C. Transmission

There are 4 transmission channels used in this robot. They are listed below -

- A 2.4 GHz receiver in the body that receives commands from a 2.4 GHz Transmitter in the station and drives the robot accordingly.
- An onboard HC-12 transceiver module is used to transmit all of the sensor data to the stations HC-12 module as a single array. It communicates in a range of 433.4-473.0 MHz.
- The FPV Wide-angle Camera and the microphone on-board transmits through a 5.8 GHz Audio Video Transmitter which is received by another 5.8GHz receiver at the station.
- The pair of walkie talkies communicate between them in a frequency range of 462.5625 MHz - 462.6625 MHz.

D. The application portal

The language of the software that was used for this project is a platform independent. But due to the drivers used for the hardware, the program only runs on a Linux platform. This section will cover the different software components of the system.

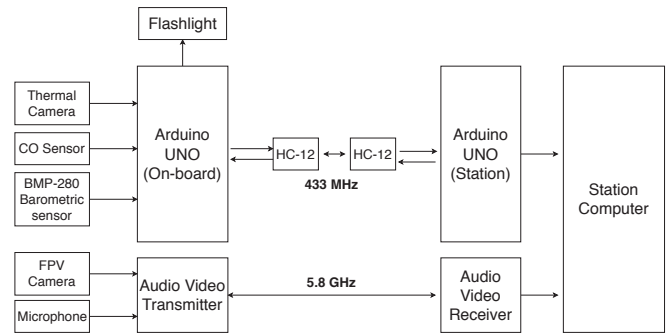


Figure 2: Flow of Data



Figure 3: Running on the debris

1) *Streaming*: For real-time video streaming, we are using the MJPEG streamer package with the V4L2 driver in Linux operating system. In the station, we are receiving the audio-video feed and sensor data using 5.8GHz receiver and HC12 accordingly. Then the video is fed to the computer using a USB TV tuner.

2) *User Interface*: The user interface for the system is shown in Figure 4b, ?? and 5b. The interface is divided into several parts: two camera displays, and an experimental parameter console. Any user can easily watch the live streaming from this user interface.

V. EXPERIMENTAL DEPLOYMENT

The robot was deployed on the first floor of an under-construction building. The floor was covered with debris, yet the robot could scale through the uneven terrain (Figure 3). It had sufficient power to run over small obstacles and could climb inclinations up to 60 degrees.

It was deployed at night to test the video and thermal camera feed. The LED flashlight in front of the bot was bright enough to illuminate the path ahead of the bot. The video feed was also very clear. The thermal camera feed was tested with 3 test cases. In the first case, a person sat in front of the bot (Figure 4). In the second case, an arm was stretched in front of the bot (Figure 5). In all three cases, the thermal camera feed could detect the outline of the part of the body that indicated that

a living being is in front of the bot. Meanwhile, the onboard sensors were continuously updating the ambient temperature, the amount of Carbon Monoxide gas in the environment and the altitude of the bot from the reference level.

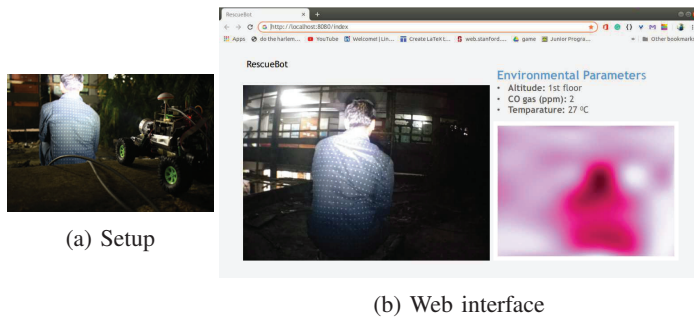


Figure 4: A person sitting in front of the camera

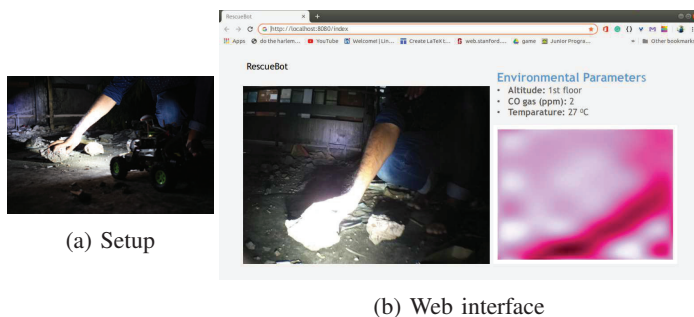


Figure 5: A person's hand in front of the camera

VI. CONCLUSION AND FUTURE WORKS

Urban search and rescue robotics is an intriguing field of study, but it is also a very challenging one. In this paper, we have described a working prototype of an urban search and rescue robot employing structural and controlling features designed to enhance the capabilities of the robot. Future works include the robustness of the system, first and foremost. Unexpected or complex situations almost always cause the robot to make mistakes, mistakes that easily could compromise a real USAR operation. USAR professionals need to be able to rely on all the equipment they use, so this poses a significant problem. This applies to both the software and the hardware parts of the project. A human detection algorithm could be implemented for the automatic detection of living humans and an elegant indoor positioning system can be employed to pinpoint the human's location. An autonomous mapping and path planning system can make the robot much easier to control and will help the operator to make fewer mistakes. An auto mode could be incorporated so that whenever the robot loses communication, it can return to the base or move to the closest area with better reception. At last, with proper modification and implementation, this prototype could handle emergency situations with ease and save significant number of lives.

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